SAMPLE

General Relativity

Gravity as Curved Spacetime

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Introduction

General relativity is Einstein's theory of gravity, and gravity, under this description, is not a force at all. It is geometry. A massive body does not reach across space and tug on other bodies; it curves the space and time around it, and everything else simply follows the straightest available path through a region that is no longer flat. What Newton wrote as an attraction between masses, Einstein wrote as a relation between matter and the shape of spacetime itself. The famous summary, offered by the relativist John Archibald Wheeler, compresses the whole theory into a sentence: spacetime tells matter how to move, and matter tells spacetime how to curve.

Einstein began the work in 1907 with a thought experiment about a person falling freely, realized that such a person would feel no gravity at all, and spent the next eight years converting that single insight into a theory of physics expressed in the language of curved geometry. He presented the final field equations to the Prussian Academy of Sciences in Berlin in November 1915, and within a year the theory had explained a decades-old mystery in the orbit of Mercury, predicted that starlight would bend around the Sun, and acquired, through the astronomer Karl Schwarzschild's solution, its first exact description of the space around a star.

The theory's reach has only grown. General relativity predicts black holes, the expanding universe, gravitational waves, the slowing of clocks in gravitational wells, and the bending, brightening, and delaying of light by every mass it passes. Its equations govern the design of satellite navigation systems, the interpretation of observations from radio telescopes and laser interferometers, and the frontiers of cosmology and quantum physics. Widely acknowledged as a theory of extraordinary mathematical beauty, it has also survived a century of increasingly precise attempts to break it.

The chapters that follow trace the theory in plain language, with no equations: its origins in a falling body and a curious mind, the geometry that made it possible, the tests that confirmed it, the objects, from black holes to gravitational waves, that it brought into physics, the cosmology built on its foundations, the unfinished search for its quantum successor, and the reasons it remains, a century on, the working theory of space, time, and gravity. It is a story with an unusual arc: a theory dismissed as untestable in its youth became, in its second century, among the most tested in all of science, and the experiments that were once its critics are now its working hands.

Chapter 1: The Falling Man and the Making of a Theory

Every account of general relativity begins in the same place, because Einstein himself put it there. In 1907, while working as a patent clerk in Bern and still refining the special theory of relativity he had published two years earlier, he sat at his desk and had what he would later describe as the happiest thought of his life: someone falling freely from a roof feels none of their own weight. While the fall lasts, gravity vanishes from the falling person's experience. Loose objects drift alongside them as though no force acted on anything.

The observation sounds trivial, but it contains a seed that took Einstein eight years to grow. The feeling of weight, it suggests, is not the direct perception of a gravitational force; it is what happens when something prevents you from following gravity's natural motion. A person standing on a floor feels weight because the floor pushes back. A person in free fall feels nothing at all because nothing intervenes. If the sensation of gravity can be abolished by falling, then perhaps gravity is not, at bottom, a force in the ordinary sense, and perhaps the natural, unforced motion of a body near the Earth is not stillness but the fall itself.

That idea had an experimental partner already in place. Astronomers and physicists had long known, with growing precision, that all bodies fall with the same acceleration regardless of what they are made of. Newton had been forced to assume this coincidence, giving inertial mass, the measure of a body's resistance to being pushed, and gravitational mass, the measure of its response to gravity, exactly the same value without any explanation for why the two should agree. The Hungarian physicist Loránd Eötvös spent decades verifying the agreement in delicate balance experiments, and found no discrepancy. Einstein took the equality not as an accident but as a clue: if the response to gravity is universal, then gravity must be a property of space and time rather than of the individual bodies moving through them.

The path from that clue to a finished theory was neither quick nor straight. Einstein tried and abandoned ideas for years, published false starts, and was repeatedly blocked by mathematics he did not yet command. What he needed, he slowly realized, was a geometry suited to curved spaces, and he found it ready-made: the non-Euclidean geometry developed by Bernhard Riemann in the nineteenth century, in which the familiar rules of flat space, parallel lines and straight lines and the angles of triangles, give way to curvature that varies from place

to place. Mathematicians in Zurich, among them Einstein's classmate Marcel Grossmann, taught him the machinery. A version of Riemannian geometry, adapted to space and time together, became the language in which the theory was finally written.

The end came in November 1915, in four weekly lectures before the Prussian Academy in Berlin. In the first, Einstein offered field equations he did not yet trust; by the eighteenth of November he had recalculated the orbit of Mercury and found his theory explaining, without adjustment, an anomaly that had troubled astronomers for half a century; and in the final lecture, on the twenty-fifth of November, he presented the equations in the form still used today. He later described the intense month as a time of exhaustion and fluctuating confidence, and the arrival at the correct equations as a moment of palpitation of the heart. The falling man of 1907 had, at last, reached the ground.